Medical Terminology

Definition

Medical terminology is language used to precisely describe the human body including its components, processes, conditions affecting it, and procedures performed upon it. Medical terminology is used in the field of medicine.

Medical terminology has quite regular morphology, the same prefixes and suffixes are used to add meanings to different roots. The root of a term often refers to an organ, tissue, or condition. For example, in the disorder hypertension, the prefix "hyper-" means "high" or "over", and the root word "tension" refers to pressure, so the word "hypertension" refers to abnormally high blood pressure. The roots, prefixes and suffixes are often derived from Greek or Latin, and often quite dissimilar from their English-language variants. This regular morphology means that once a reasonable number of morphemes are learnt it becomes easy to understand very precise terms assembled from these morphemes. A lot of medical language is anatomical terminology, concerning itself with the names of various parts of the body.

Basics

There are three basic parts to medical terms:

1. **Root:** the middle of the word and its central meaning.
2. **Prefix:** comes at the beginning and usually identifies some subdivision or part of the central meaning.
3. **Suffix:** comes at the end and modifies the central meaning as to what or who is interacting with it or what is happening to it.

Examples of Prefixes

Component	Meaning	Example
a/an	without, none	anemia (literally no blood but means few red cells)
Micro	small	microstomia (abnormally small mouth)
macro	large	macrostomia (abnormally large mouth)

Component	Meaning	Example
mega	enlarged	megacolon (abnormally large colon)

Examples of Suffixes

Component	Meaning	Example
-itis	inflammation	Carditis
-osis	abnormal condition	cyanosis (blue color of the skin and mucous membranes)
-ectomy	to cut out (remove)	tonsillectomy
-otomy	to cut into	tracheotomy (to cut into the trachea temporary opening)
-ostomy	to make a "mouth"	colostomy (to make a permanent opening in colon)
-scopy/-scopic	to look, observe	colonoscopy (look into colon)
-megaly	abnormal large	Cardiomegaly
-graphy/-graph	recording an image	Mammography (procedure) / Mammograph (device)
-gram	the image (X-ray)	Mammogram
-ology	science	Biology
-ologist	specialize in	Cardiologist

Prefix and Suffix Changes (Example: Myocarditis)

- **myo** = muscle (prefix)
- **card** = heart (root)
- **itis** = inflammation (suffix)

> - **Myocarditis** = muscle layer of heart inflamed
> - **Pericarditis** = outer layer of heart inflamed
> - **Endocarditis** = inner layer of heart inflamed
> - **Cardiologist** = a physician specializing in the heart

- **Cardiomegaly** = enlargement of the heart
- **Carditis** = inflammation of the heart

Digestive System Terms

Component	Meaning	Example
Gastr/o	Stomach	Gastritis, Gastrectomy
Hepat/o	Liver	Hepatitis, hepatoma
Chole	Gall, bile	Cholecystitis, cholecystectomy
Cyst/o	Bladder, sac	Cystitis
Emes/o	Vomit	Emesis, emetic, antiemetic
Lapar/o	Abdominal wall	Laparotomy
-rrhea	Flow, discharge	Diarrhea
-phagia	Ingestion/swallowing	Dysphagia
itis	Inflammation	Peritonitis, Pharyngitis, Oesophagitis, Gastritis, Hepatitis

The Digestive System

The digestive system consists of:

- The mouth cavity
- Salivary glands
- Pharynx
- Esophagus
- Stomach
- Small intestines
- Large intestines
- Liver and biliary glands
- The pancreas

1. Mouth Cavity & Salivary Glands

Contents of the mouth: Tongue, teeth and the openings of the salivary glands.

Teeth: There are two types of teeth:

- **Milk (deciduous) teeth:** Temporary teeth present in the children, twenty in number ten above and ten below.
- **Permanent teeth:** They are thirty two teeth, sixteen above and sixteen below.

The Tongue: is a muscular organ covered by mucous membrane. It consists of three parts: tip, root and body. It is responsible for speech, taste and swallowing.

Salivary Glands: There are three types of salivary glands:

- **Parotid gland:** It is the largest salivary gland.
- **Submandibular gland:** It is about half the parotid size, lies deep to and below the mandible.
- **Sublingual gland:** It is the smallest salivary gland, lies under the mucous membrane of the floor of the mouth called sublingual fold.

2. Pharynx

- **Nasopharynx:** It lies behind the nasal cavity, above the soft palate and acts as a passage for air only.
- **Oropharynx:** It lies behind the oral (mouth) cavity, below the soft palate and acts as a passage for food and air.
- **Laryngopharynx:** It lies behind the larynx and acts as a passage for food only.

3. Oesophagus

It is a muscular tube twenty five cm long. It begins in the neck as continuation of the pharynx. The esophagus has three parts; short cervical part, long thoracic part and very short abdominal part.

4. Stomach

It lies in the upper part of the abdominal cavity behind the left lobe of liver and anterior wall of abdomen. It lies in front of the left. It is the most dilated part of the digestive tract.

5. Small Intestine

- **Duodenum:** The shortest and widest part of the small intestine. It is C-shaped tube, twenty five cm in length with its concave part directed to the left side and is occupied by the head of the pancreas.
- **Jejunum**
- **Ileum**

6. Large Intestine

- **Caecum:** It lies in the lower right part of the abdominal cavity. It has lower blind end and it is continuous with ascending colon. The lower end of the ileum and the appendix open into it.
- **Vermiform appendix:** It is variable in length and in position and it opens into the caecum below the opening of the ileum.
- Ascending colon, Right colic flexure, Transverse colon, Left colic flexure, Descending colon, Sigmoid (pelvic) colon, Rectum, Anal canal which opens below by the anus.

7. Pancreas

It consists of four parts: head, neck, body and tail.

- **Endocrine function:** It secretes two hormones which regulate the level of the blood glucose:
    - **Insulin hormone:** lowers the blood glucose level.
    - **Glucagon hormone:** Elevates the blood glucose level.
- **Exocrine function:** it secretes pancreatic enzymes in the pancreatic juice which helps the digestion of food. The pancreatic juice is secreted through the pancreatic duct which unites with the bile duct and open together into the duodenum.

Hospitals and Healthcare

Hospital is an institution that provides medical services for the community. The doctors, nurses and other personnel of hospital work to restore health to sick and injured people. They also try to prevent diseases and maintain health throughout the community. In addition, a hospital services as a center for medical education and research.

- In the industrial world, a hospital is a clean building or complex equipped with modern medical facilities where doctors and nurses use up to date techniques to treat patients.
- In some developing countries, hospitals may be nothing more than crude huts or tents, where doctors and nurses work with limited drug supplies and medical equipment.
- Experts usually measure the level of the hospital in terms of the number of the hospital bed it has available for patients.

Hospital Wards and Units

Millions of people must, at some point in their lives, spend sometimes in hospitals as inpatients. Millions of others will visit hospital as outpatients. They receive treatment but don't stay at hospital.

Most patients who are admitted to hospital are nursed on either a medical ward or a surgical ward.

- **Medical wards** cater for illnesses such as gastrointestinal diseases.
- The majority of patients in **surgical wards** have an operation during their stay in hospital, and different nursing care is needed.

Special Wards or Units include:

- Pediatric unit for children
- E.N.T. ward (ear, nose, throat)
- Ophthalmic ward (eyes)
- Orthopedic ward (bones)
- Psychiatric unit (mental illness)
- Geriatric Ward
- Dermatological Ward
- Intensive Care Unit
- X-ray Department
- Central Sterile Supply Department
- Laboratory
- Antenatal Clinic
- Maternity Unit
- Occupational Therapy
- Physiotherapy
- Infectious Diseases Unit

Nurses

Nurses play a vital role. The nursing staff forms the largest group on the patient care team.

- Professional nurses are those who have graduated from a school or college of nursing.
- They carry out much of the patients' care under the direction of doctors.
- Nurses also direct other members of the nursing staff, including nursing auxiliaries.
- These men and women do many routine necessary tasks and free the nurses for work requiring their special skills.

CPR & Emergency Medical Response

What is CPR?

Cardiopulmonary Resuscitation is an emergency lifesaving procedure performed when the heart stops beating. Immediate CPR can INCREASE chances of survival after cardiac arrest.

Why Should We Learn CPR?

The nature of your job is to be prepared to handle medical emergencies. You must believe that knowing the skills of CPR makes you a more useful member of your community. Whatever your reasons, it is important to remember that CPR can help to save lives. Your hard work and study can make a difference.

Definitions of Death

- **Clinical death:** means that the heartbeat and breathing have stopped. This is best thought of as near or apparent death, and it may be averted, or reversed. ("Sudden death" is sudden, unexpected clinical death).
- **Biological death:** is permanent brain death due to lack of oxygen. This death is final.

Causes of sudden death include: Suffocation, Drowning, Electrocution & drug overdose, Severe allergic reaction & car accidents.

Many of these deaths can be prevented if the victims get prompt help...If someone trained in CPR provides proper lifesaving measures until trained professionals take over.

When CPR is started within 4 minutes, the victim's chances of leaving the hospital alive are four times greater than those of victim who does not receive CPR until after 4 minutes. During the first few minutes of clinical death, promptly initiated CPR may turn the victim back to life. Without CPR, biological death will occur. Speed in starting CPR and in getting specialized medical care for the victim are the keys to saving lives. **Speed is important. Immediate CPR is CRITICAL.**

What is Cardiac Arrest vs Heart Attack?

- **Cardiac Arrest:** Occurs when the heart malfunctions and stops beating unexpectedly. Cardiac arrest is an **"ELECTRICAL"** problem.

- **Heart Attack:** Occurs when blood flow to the heart is blocked. A heart attack is a **"CIRCULATION"** problem.

About two-thirds of deaths from heart attack occur before the Victim reaches the hospital.

5 Reasons You Should Learn CPR

1. **It's Easy:** CPR doesn't require years of medical training to learn. It most often takes a single class through a certified organization.
2. **Increased Survival Rate:** There are more than 300,000 out-of-hospital cardiac arrests annually in the US, and nearly 90% of them are fatal. Performing CPR provides oxygen to the brain and other vital organs to give the patient the best chance of recovery.
3. **Increased Confidence:** With CPR training, you gain the skills and knowledge necessary to perform CPR with confidence, and to make good decisions that can help someone in distress.
4. **Save Someone You Love:** Almost 70% of Sudden Cardiac Arrests (SCA) occur at home. You're more likely to use your CPR skills to help someone close to you. It affects people of all ages, including children and teens.
5. **You Get to Use an AED:** AED's (automatic electronic defibrillator) training is often included. They are portable machines used to deliver an electric shock through the chest to the heart, to re-establish a normal heart rhythm.

Evaluation

1. Name the common causes of sudden death?
2. Define the clinical death?
3. Define the biological death?
4. When does the brain damage begin?
5. When the brain death is certain?
6. CPR must start when....